

Saumik Dana

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Summary

US green card holder who builds and operates live ML- and LLM-driven systems end-to-end — research, deployment, and infrastructure as one workflow. PhD from The University of Texas at Austin.

Experience

Quantitative Researcher, Zebra Capital Management LLC, Stamford CT, **Feb 2024 – Dec 2025**

ML-driven trading

- Built live intraday options strategies on Bayesian anomaly-detection signals, with stochastic volatility calibration via FFT options pricing; weekly PnL of 5–10% of peak capital at risk
- Deployed live equity portfolio: per-ticker α/β predictions $\rightarrow$ NSGA-II Pareto-optimal weights, HMM regime detection, Jensen–Shannon rebalancing; 5%+ annualized alpha on multi-year backtest
- Built multi-asset interday options strategies: TabPFN classifiers on backtested PnL, with LoRA-adapted Chronos embeddings of technical indicators as features

LLM-driven trading

- Ran intraday options trading on LLM reasoning (gpt-oss-120b) over stochastic volatility signals: state-dependent prompts with verdict history and entry-thesis read-back, versioned and evaluated over fixed windows
- Deployed intraday FX system synthesizing news and price-action narratives via CoT scaffolding (llama-3.1-8b), with DSPy (GEPA) optimization over archived live runs scored against market prices
- Deployed intraday commodity system reasoning over macro news and sovereign yield-curve differentials, with evidence attribution, conflict-resolution, and abstention rules

LLM agents

- Built MCP-wrapped tool-calling pipeline mining trade signals against backtested PnL — surfaced 1.5+ Sharpe strategies
- Built multimodal PDF QA system (Qwen2.5-VL) with contrastive-embedding fine-tuning and Qdrant indexing

Infrastructure

- Designed serverless AWS (Lambda, EventBridge, DynamoDB, S3, CloudFormation) and Modal stacks; tuned Lambda for ML inference with S3-backed HuggingFace/Torch caches
- Engineered GitHub Actions CI/CD with digest-pinned ECR images and DST-aware scheduling; built React dashboards over DynamoDB/S3 via FastAPI

Computational Engineer, VISIE Inc., Austin TX, **Aug 2023 – Nov 2023**

- Implemented TCP/UDP motion-control protocols for the robotic arm in a surgical navigation platform
- Supported packaging, deployment, and the live demonstration that anchored a successful Series A raise

Computational Lead, Sophelio, Austin TX, **Aug 2022 – Mar 2023**

- Adapted physics-informed modeling from fusion data to financial time series: sparse-regression PDE construction and signal generation
- Implemented CAGR-maximizing Bayesian TPE-optimized swing trading system deployed on AWS Lambda

Education

PhD, Engineering Mechanics — University of Texas at Austin, Austin TX

ME, Mechanical Engineering — Indian Institute of Science, Bangalore, India

BE, Mechanical Engineering — University of Mumbai, Mumbai, India

Technical Skills

ML/Quant — PyTorch, scikit-learn, TabPFN, Chronos, PEFT/LoRA, hmmlearn, SciPy, pymoo, autograd

LLM — Groq, LangChain, Fireworks AI, Qdrant, DSPy, MCP, tool calling

Cloud — AWS (Lambda, DynamoDB, S3, ECR, EventBridge, API Gateway, CloudFormation), Modal, Docker

Data/APIs — Python, pandas, Polars, NumPy, FastAPI, React, Next.js, Alpaca, OANDA, TradeStation, yfinance